

The fault-tolerant bipancyclicity of the even k -ary n -cube

Jywe-Fei Fang, Chen-Hsiang Liao, Kuo-Hung Tseng and Ming-Yan Chang

Department of Digital Content and Technology

National Taichung University

jffang@mail.ntcu.edu.tw

Abstract

The interconnection network considered in this paper is the k -ary n -cube that is an attractive variance of the well-known hypercube. Many interconnection networks, including the ring, the torus and the hypercube, that are attractive in both theoretical interests and practical systems can be regarded as the subclasses of the k -ary n -cubes. In this paper, we investigate the fault tolerant cycle embedding properties of the k -ary n -cubes. We show that the k -ary n -cube is $(2n-2)$ -edge-fault bipancyclic for k is even and $n \geq 2$. That is, it embeds all cycles of even lengths ranging from 4 to N , where N is the order of the network when there exists at most $2n-2$ edge faults. Since the degree of the k -ary n -cube is $2n$, the result is optimal.

1 Introduction

In massively parallel systems, the selection of an interconnection network is crucial as it can greatly affect communication performance, hardware costs, potentialities for efficient algorithms and fault tolerant capabilities, among others [11]. An interconnection network is usually represented by a graph where the vertices represent the nodes and the edges represent the links. Various interconnection networks have been proposed; thus the portability of algorithms across these interconnection networks is of considerable importance. That a host interconnection network can embed a guest interconnection network implies that the algorithms on the guest can be simulated on the host systematically.

Paths and *cycles* are popular interconnection networks due to their simple structures and low degrees. That is why many parallel algorithms have been devised based on them [13, 16]. Many researchers have discussed how to embed cycles and paths into various topologies [8, 13, 20]. A graph is *Hamiltonian* if it embeds a Hamiltonian cycle that contains each vertex exactly once [13]. In other words, that a graph is Hamiltonian implies that it embeds the maximal cycle. To execute a parallel program efficiently, the length of the allocated cycle must correlate with the

problem size of the program. For this reason, many researchers have proposed various methods for embedding cycles of different lengths into an interconnection network. A graph is *pancyclic* if it embeds cycles of all lengths ranging from 3 to N , where N is the order of the graph [2]. A graph is *m-pancyclic* if it embeds cycles of all lengths ranging from m to N , where $3 \leq m \leq N$.

Fault tolerance is an important property of an interconnection network because vertex faults and edge faults may occur on it. Various fault tolerant properties have been investigated in many interconnection networks [5-7, 21]. A graph G is *k-edge-fault tolerant Hamiltonian* graph if $G(V, E-F)$ remains Hamiltonian for an arbitrary faulty edge set $F \subset E$, where $|F| \leq k$. A graph G is *k-edge-fault tolerant pancyclic* graph if $G(V, E-F)$ remains pancyclic for an arbitrary faulty edge set $F \subset E$, where $|F| \leq k$. The *k-edge-fault tolerant Hamiltonicity* and *k-edge-fault tolerant pancyclicity* have been widely studied in many interconnection networks [9, 12, 19].

The interconnection network considered in this paper is the *k-ary n-cube* which is denoted by a $Q(k, n)$. Many interconnection networks, including the *ring*, the *torus* and the *hypercube*, that are attractive in both theoretical interests and practical systems can be regarded as the subclasses of the *k-ary n-cubes* [10, 11]. In fact, they are widely applied as the interconnection networks of some practical systems. For example, Kendall square machine has ring structure [10], the Tera Parallel Computer [18] and CRAY T3D [15] use the 2D torus and the 3D torus as their interconnection networks, respectively. The Symult S-series [1], NCUBE family [14] and the Cosmic Cube [17] employ the hypercube as their interconnection networks.

On the other hand, the $Q(k, n)$ has been proved to possess many attractive properties such as *regularity*, *vertex transitivity* and *edge transitivity* [3]. For example, Bose et al. shown that it is Hamiltonian; and they proposed a single-vertex routing algorithm and a broadcasting algorithm [4]. Ashir et al. devised several communication algorithms including multi-vertex broadcasting, single-vertex scattering and total exchange [3]. Yang et al. investigated the fault

tolerant Hamiltonicity [21].

In this paper, We do a further investigation about the fault-tolerant bipancyclic properties of the $Q(k, n)$. We refer to the $Q(k, n)$ for k is even (respectively, odd) as the *even* (respectively, *odd*) $Q(k, n)$. We show that the even $Q(k, n)$ is $(2n-2)$ -edge-fault bipancyclic for $n \geq 2$. That is, it embeds all cycles of even lengths ranging from 4 to N with at most $2n-2$ edge faults. Owing to that the connectivity of the $Q(k, n)$ is $2n$ [4], our result is optimal.

2 Notations and background

A path of the length s is denoted by P_s ; and a cycle of the length s is denoted by C_l where $l \geq 3$. A ladder of the length s , denoted by L_s , is a $P_s \times K_2$, where K_2 is a two-vertex complete graph. Each vertex of L_s is labeled by (b_2, b_1) , where $b_1 = 0$ or $b_1 = 1$, and $0 \leq b_2 \leq s$. Each edge $((b_2, 0), (b_2, 1))$ is called a *rung* of L_s , where $0 \leq b_2 \leq s$. Specifically, it is called the b_2 th rung. The 0th rung is called the *bottom rung* of the ladder; whereas the s th rung is called the *top rung* of the ladder. In this paper, we use $[(0, 0), (1, 0), \dots, (s, 0), (s, 1), \dots, (1, 1), (0, 1)]$ to denote L_s . Clearly, a path of the length $2l+1$, $((0, 0), (1, 0), \dots, (l, 0), (l, 1), \dots, (1, 1), (0, 1))$, can be embedded in L_s , where $0 \leq l \leq s$.

Proposition 1. There exists a path of each odd length ranging from 1 to $2s+1$ joining $(0, 0)$ and $(0, 1)$ in L_s . Thus, there exists a cycle of each even length ranging from 4 to $2s+2$ containing $((0, 0), (0, 1))$.

Definition 0. A torus with R rows and F columns, denoted by $TOR(R, F)$, is defined as $C_R \times C_F$.

A vertex of $TOR(R, F)$ is labeled by (v_2, v_1) , where $0 \leq v_2 \leq R-1$, $0 \leq v_1 \leq F-1$. The cycle, $((v_2, 0), (v_2, 1), \dots, (v_2, F-1), (v_2, 0))$, is denoted by $(v_2, *)$; and the cycle, $((0, v_1), (1, v_1), \dots, (R-1, v_1), (0, v_1))$, is denoted by $(*, v_1)$.

Definition 1. The k -ary n -cube, denoted by the $Q(k, n)$, is defined recursively [4]:

1. $Q(k, 1)$ is C_k .
2. $Q(k, n)$ is $Q(k, n-1) \times C_k$ for $n \geq 2$.

That is, a $Q(k, n)$ is a C_k^n . $Q(k, n)$ comprises k^n vertices, each vertex x labeled by an n -digit number in radix k arithmetic $v_n v_{n-1} \dots v_2 v_1$. The vertex $x = v_n v_{n-1} \dots v_{i+1} v_i v_{i-1} \dots v_2 v_1$ is adjacent to another vertex $y = u_n u_{n-1} \dots u_{i+1} u_i u_{i-1} \dots u_2 u_1$ if and only if they differ by exactly one digit position i and $|v_i - u_i| = 1$, where $1 \leq i \leq n$ and (x, y) is called the dimension i edge. A vertex $x = v_n v_{n-1} \dots v_{i+1} v_i v_{i-1} \dots v_2 v_1$ is an *even* (respectively, *odd*) vertex if the sum of its all digits is even (respectively, odd). The even $Q(k, n)$ is a bipartite graph because the

odd (respectively, even) vertices are just adjacent to even (respectively, odd) vertices.

A path of $Q(k, n)$ can be represented by its *transition sequence* which is the ordered list of each digit position associated with the direction (i.e., + or -) that changes as it proceeds from one vertex to the next one. For example, the path of $Q(5, 4)$, (2314, 2324, 2224, 3224, 3223, 3213), can be represented by the transition sequence (2+, 3-, 4+, 1-, 2-). $Q(k, n)$ possesses many attractive properties:

Proposition 2. $Q(k, n)$ is Hamiltonian [4].

Proposition 3. $Q(k, n)$ is vertex transitive and edge transitive [3].

Proposition 4. $Q(2, n)$ is $(2n-2)$ -edge-pancyclic [22].

Proposition 4.1. There exists a $(2n-2)$ -fault Hamiltonian cycle in the even $Q(k, n)$ [21].

The vertex $x = v_n v_{n-1} \dots v_2 d$ is called the vertex d of the $V^* = v_n v_{n-1} \dots v_2^*$. The *outline graph* of $Q(k, n)$ with edge faults, denoted by $OG(Q(k, n))$, is to take each $v_n v_{n-1} \dots v_2^*$ subnetwork as a supervertex, where $*$ is a don't care symbol; and the supervertices V^* is adjacent to the supervertices U^* in $OG(Q(k, n))$ if and only if Vd is adjacent to Ud for each $0 \leq d \leq k-1$. Clearly, $V^* = v_n v_{n-1} \dots v_2^*$ is adjacent to $U^* = u_n u_{n-1} \dots u_2^*$ if and only if they differ by exactly one digit position, the i th digit, where $2 \leq i \leq n$, $|v_i - u_i| = 1$, and (Vd, Ud) is non-faulty for each $0 \leq d \leq k-1$. That is, if each $v_n v_{n-1} \dots v_2^*$ subnetwork of $Q(k, n)$ is taken as a supervertex, $Q(k, n)$ with f_1 faulty edge faults will be transformed into $Q(k, n-1)$ with f_2 edge faults, where $f_2 \leq f_1$. We have the following proposition.

Proposition 5. $OG(Q(k, n))$ is $Q(k, n-1)$ with f_2 edge faults, where $Q(k, n)$ has f_1 edge faults, $f_2 \leq f_1$.

By the structure of the $Q(k, n)$, the vertex d of V^* and the vertex d of U^* are adjacent for each $0 \leq d \leq k-1$ if V^* and U^* are adjacent in $OG(Q(k, n))$. Clearly, we have:

Proposition 6. If $OG(Q(k, n))$ embeds C_l , $(V_0^*, V_1^*, V_2^*, \dots, V_{l-1}^*, V_l^*)$, $Q(k, n)$ embeds the structure of $C_l \times C_k$ (i.e., $TOR(l, k)$), where the cycles $(*, v_1)$ contain no edge faults, where $0 \leq v_1 \leq k-1$.

Likewise, if $OG(Q(k, n))$ embeds P_l , $(V_0^*, V_1^*, V_2^*, \dots, V_{l-1}^*, V_l^*)$, $Q(k, n)$ embeds the structure of $P_l \times C_k$.

As illustrated in Figure 1, the structure of $C_l \times C_k$ is embedded in $Q(k, n)$ if C_l is embedded in $OG(Q(k, n))$.

Definition 2. A *cycle-of-ladders* $COL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ is unified by a *bone cycle* BC and r ladders $LD(0), LD(1), \dots, LD(r-1)$ with $BR(0), BR(1), \dots, BR(r-1)$ as the bottom

rungs, respectively, such that each $BR(i)$ is contained in BC , where $0 \leq i \leq r-1$. From Proposition 1, we have:

Proposition 7. A cycle-of-ladders $COL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ contains a cycle of each length l ranging from $|BC|$ to N and $l-|BC|$ is even, where the length of BC is denoted by $|BC|$.

Definition 4. A cycle-of-ladders with only faulty rungs except the bottom rungs and the top rungs is called a *ladder-wounded cycle-of-ladder (LWCOL for short)*.

In $LWCOL(BC, r, LD(0), LD(1), \dots, LD(r-1))$, a *block* of faulty rungs with the length f_s is a set of successive f_s faulty rungs. The *ladder block length* of $LD(i)$ is the length of the maximum block that locates in $LD(i)$, where $0 \leq i \leq r-1$. For example, consider $LWCOL(BC, 3, LD(0), LD(1), LD(2))$ in Figure 2. The ladder block lengths of $LD(0)$, $LD(1)$ and $LD(2)$ are 0, 2 and 1, respectively. The *LWCOL block length* of $LWCOL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ is the length of the maximum block that locates in each $LD(i)$, where $0 \leq i \leq r-1$. The *LWCOL block length* of $LWCOL(BC, 3, LD(0), LD(1), LD(2))$ in Figure 2 is 2.

3 The results

Consider $LWCOL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ with one non-faulty ladder $LD(0)$ of the length p_0 and the *LWCOL block length* f_s , where $f_s \leq p_0$. Let the length of the ladder $LD(i)$ be p_i , where $1 \leq i \leq r-1$. Clearly, $LWCOL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ embeds $LWCOL(BC, 2, LD(0), LD(1))$. If the j th rung in $LD(1)$ is non-faulty, $LWCOL(BC, 2, LD(0), LD(1))$ embeds a cycle of each length l ranging from $|BC|+2j$ to $|BC|+2j+2p_0$ and $l-|BC|$ is even. Therefore, it embeds a cycle of each length l ranging from $|BC|$ to $|BC|+2p_0+2p_1$ and $l-|BC|$ is even because $f_s \leq p_0$; and the bottom rung and the top rung are non-faulty from definition. We can append all vertices of each $LD(i)$ one by one to the bone cycle BC for each $1 \leq i \leq r-1$ because the top rungs are non-faulty. Thus, we can prove that $LWCOL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ contains a cycle of each length l ranging from $|BC|+2(p_0+p_1+\dots+p_{i-1}+p_i)$ to $|BC|+2(p_0+p_1+\dots+p_i+p_{i+1})$ and $l-|BC|$ is even for each $1 \leq i \leq r-2$. Thus, we have:

Lemma 1. $LWCOL(BC, r, LD(0), LD(1), \dots, LD(r-1))$ with one non-faulty ladder $LD(0)$ of the length p_0 and the *LWCOL block length* f_s contains a cycle of each length l ranging from $|BC|$ to N and $l-|BC|$ is even, where $f_s \leq p_0$.

For example, $LWCOL(BC, 3, LD(0), LD(1), LD(2))$ in Figure 2 embeds a cycle of each even

length ranging from 6 to 24.

Lemma 2. The even $TOR(k, k)$ (i.e., $Q(k, 2)$) with at most two edge faults is bipancyclic for $k \geq 6$.

Proof. By the symmetric properties of the even $TOR(k, k)$, without loss of generality, let an edge fault be $(x_1, x_2) = ((0, 0), (0, 1))$.

Case 1: The other edge fault $(y_1, y_2) = ((v_2, v_1), (v_2, v_1+1))$. Without loss of generality, we assume that $k/2 \geq v_2 \geq 0$ and $k/2 > v_1 \geq 0$. There exists a non-faulty ladder $[(0, k-2), (1, k-2), \dots, (k-2, k-2), (k-1, k-2), (k-1, k-1), (k-2, k-1), \dots, (1, k-1), (0, k-1)]$. From Proposition 1, there exists a cycle of each even length ranging from 4 to $2k$.

Subcase 1.1: $v_1 \neq 0$ and $v_2 \neq 0$. Clearly, there exists u_2 such that $\{u_2, u_2+1\} \cap \{0, v_2\} = \emptyset$. Let $BC = ((u_2+1, 0), (u_2+1, 1), \dots, (u_2+1, k-2), (u_2+1, k-1), (u_2+1, 0))$, $LD(0) = [(u_2+1, 0), (u_2+2, 0), \dots, (u_2+k-1, 0), (u_2, 0), (u_2, 1), (u_2+k-1, 1), \dots, (u_2+1, 1)]$, $LD(1) = [(u_2+1, 2), (u_2+2, 2), \dots, (u_2+k-1, 2), (u_2, 2), (u_2, 3), (u_2+k-1, 3), \dots, (u_2+1, 3)]$, $\dots$, $LD(k/2-1) = [(u_2+1, k-2), (u_2+2, k-2), \dots, (u_2+k-1, k-2), (u_2, k-2), (u_2, k-1), (u_2+k-1, k-1), \dots, (u_2+1, k-1)]$, where all the first parameters are positive and expressed modulo k . Unifying the bone cycle BC with the $k/2$ ladders, $col = COL(BC, k/2, LD(0), LD(1), \dots, LD(k/2-1))$ can be derived. The length of each $LD(i)$ is $k-1$, where $0 \leq i \leq k/2-1$. There exists a non-faulty ladder $LD(i)$ because there are at least 3 ladders for $k \geq 6$. The faulty rungs $((0, 0), (0, 1))$ and $((v_2, v_1), (v_2, v_1+1))$ are neither bottom rungs nor top rungs. From Lemma 1, there exists a cycle of each even length ranging from k to $k+2(k-1)k/2 = k^2$.

Subcase 1.2: $v_2 = 0$. The proof is similar to the one of Subcase 1.

Subcase 1.3: $v_1 = 0$. The proof is similar to the one of Subcase 1.

Case 2: The other edge fault $(y_1, y_2) = ((v_2, v_1), (v_2+1, v_1))$. Without loss of generality, we assume that $k/2 > v_2 \geq 0$ and $k/2 \geq v_1 \geq 0$. Similar to Case 1, there exists a cycle of each even length ranging from 4 to $2k$.

Subcase 2.1: $v_2 = 0$ and $0 \leq v_1 \leq 1$. Without loss of generality, we assume that $v_1 = 0$. In the first stage, let $BC_1 = ((0, 1), (0, 2), \dots, (0, k-2), (0, k-1), (1, k-1), (1, k-2), \dots, (1, 2), (1, 0))$, $LD_1(0) = [(1, 2), (2, 2), \dots, (k-2, 2), (k-1, 2), (k-1, 3), (k-2, 3), \dots, (2, 3), (1, 3)]$, $LD_1(1) = [(1, 4), (2, 4), \dots, (k-2, 4), (k-1, 4), (k-1, 5), (k-2, 5), \dots, (2, 5), (1, 5)]$, $\dots$, $LD_1(k/2-2) = [(1, k-2), (2, k-2), \dots, (k-2, k-2), (k-1, k-2), (k-1, k-1), (k-2, k-1), \dots, (2, k-1), (1, k-1)]$. Unifying the bone cycle BC_1 with the $k/2-1$ ladders, $col_1 = COL(BC_1, k/2-1, LD_1(0), LD_1(1), \dots, LD_1(k/2-2))$ can be derived. From

Proposition 7, we know that there exists a cycle of each even length ranging from $2k-2$ to $2k-2+2(k-2)(k/2-1) = k^2-2k+2$. As illustrated in Figure 3, the case that $(y_1, y_2) = ((v_2, v_1), (v_2+1, v_1)) = ((0, 0), (1, 0))$ and $k = 8$ is shown. In the second stage, let BC_2 be the maximal cycle containing all vertices in col_1 , $LD_2(0) = [(2, 2), (2, 1), (2, 0), (3, 0), (3, 1), (3, 2)]$, $LD_2(1) = [(4, 2), (4, 1), (4, 0), (5, 0), (5, 1), (5, 2)]$, ..., $LD_2(k/2-2) = [(k-2, 2), (k-2, 1), (k-2, 0), (k-1, 0), (k-1, 1), (k-1, 2)]$. Unifying the bone cycle BC_2 with the $k/2-1$ ladders, $col_2 = COL(BC_2, k/2-1, LD_2(0), LD_2(1), \dots, LD_2(k/2-2))$ can be derived. From Proposition 7, we know that there exists a cycle of each even length ranging from k^2-2k+2 to $k^2-2k+2+4(k/2-1) = k^2-2$. On the hand, there exists a cycle of the length k^2 ,

$((0, 0), (0, k-1), (0, k-2), \dots, (0, 2), (0, 1),$

$(1, 1), (1, 0), (1, k-1), \dots, (1, 3), (1, 2),$

$\dots,$

$(k-2, k-2), (k-2, k-3), (k-2, k-4), \dots, (k-2, 0), (k-2,$
 $k-1),$

$(k-1, k-1), (k-1, k-2), (k-1, k-3), \dots, (k-1, 1), (k-1,$
 $0), (0, 0)).$

Thus, there exists a cycle of each even length ranging from k^2-2k+2 to $k^2-2k+2+4(k/2-1) = k^2$. Combing the above results, there exists a cycle of each even length ranging from 4 to k^2 .

Subcase 2.2: $v_2 = 0$ and $2 \leq v_1 \leq k/2$. Let $BC = ((0, 1), (0, 2), \dots, (0, k-1), (0, 0), (1, 0), (1, k-1), \dots, (1, 2), (1, 1)), LD(0) = [(1, 1), (2, 1), \dots, (k-2, 1), (k-1, 1), (k-1, 2), (k-2, 2), \dots, (2, 2), (1, 2)]$, $LD(1) = [(1, 3), (2, 3), \dots, (k-2, 3), (k-1, 3), (k-1, 4), (k-2, 4), \dots, (2, 4), (1, 4)]$, ..., $LD(k/2-1) = [(1, k-1), (2, k-1), \dots, (k-2, k-1), (k-1, k-1), (k-1, 0), (k-2, 0), \dots, (2, 0), (1, 0)]$. Unifying the bone cycle BC with the $k/2$ ladders, $col = COL(BC, k/2, LD(0), LD(1), \dots, LD(k/2-1))$ can be derived. From Proposition 1, we know that there exists a cycle of each even length ranging from $2k$ to $2k+2(k-2)k/2 = k^2$. Combining the result with the result in Case 2, there exists a cycle of each even length ranging from 4 to k^2 .

Subcase 2.3: $v_2 \neq 0$. Let $BC = ((v_2, 0), (v_2, 1), \dots, (v_2, k-2), (v_2, k-1), (v_2+1, k-1), (v_2+1, k-2), \dots, (v_2+1, 1), (v_2+1, 0)), LD(0) = [(v_2+1, 0), (v_2+2, 0), \dots, (v_2+k-2, 0), (v_2+k-1, 0), (v_2+k-1, 1), (v_2+k-2, 1), \dots, (v_2+2, 1), (v_2+1, 1)]$, $LD(1) = [(v_2+1, 2), (v_2+2, 2), \dots, (v_2+k-2, 2), (v_2+k-1, 2), (v_2+k-1, 3), (v_2+k-2, 3), \dots, (v_2+2, 3), (v_2+1, 3)]$, ..., $LD(k/2-1) = [(v_2+1, k-2), (v_2+2, k-2), \dots, (v_2+k-2, k-2), (v_2+k-1, k-2), (v_2+k-1, k-1), (v_2+k-2, k-1), \dots, (v_2+2, k-1), (v_2+1, k-1)]$, where all the first parameters are positive and expressed modulo k . Unifying the bone cycle BC with the $k/2$ ladders, $col =$

$COL(BC, k/2, LD(0), LD(1), \dots, LD(k/2-1))$ can be derived. From Proposition 7, we know that there exists a cycle of each even length ranging from $2k$ to $2k+2(k-2)k/2 = k^2$. Combining the result with the result in Case 2, there exists a cycle of each even length ranging from 4 to k^2 . Q. E. D.

Lemma 3. The even $Q(k, n)$ is $(2n-2)$ -edge-pancyclic for $k \geq 6$ and $n \geq 3$.

Proof. Consider an even $Q(k, n)$ with $2n-2$ edge faults. By the symmetric properties of the $Q(k, n)$, without loss of generality, we assume that $\lambda(n) \leq \lambda(n-1) \leq \dots \leq \lambda(2) \leq \lambda(1)$, where the number of faulty edges in dimension i is denoted by $\lambda(i)$. Clearly, $2n-2 \geq \lambda(1) \geq 2$; thus, $OG(Q(k, n))$ is $Q(k, n-1)$ with at most $2n-4 = 2(n-1)-2$ edge faults. From Proposition 4.1, $OG(Q(k, n))$ embeds a Hamiltonian cycle $(V_0^*, V_1^*, \dots, V_{N-3}^*, V_{N-2}^*, V_{N-1}^*)$, where $N = k^{n-1}$. From Proposition 6, the $Q(k, n)$ embeds a torus T that is $TOR(k^{n-1}, k)$ and the $(*, v_1)$ cycles don't contain faulty edges and the at most $2n-2$ edge faults locate in the $(v_2, *)$ cycles (i.e., the $C(k)$'s), where $0 \leq v_2 \leq k^{n-1}-1$ and $0 \leq v_1 \leq k-1$. From the $(0, *)$ cycle, we can partition the k^{n-1} $(v_2, *)$ cycles into $k^{\lfloor (n-1)/2 \rfloor}$ sections in which each section contains $k^{\lfloor (n-1)/2 \rfloor}$ successive $(v_2, *)$ cycles. That is, the i th section contains $(i\delta, *)$, $((i\delta+1, *)$, ..., $((i+1)\delta-2, *)$ and $((i+1)\delta-1, *)$, where $0 \leq i \leq k^{\lfloor (n-1)/2 \rfloor}-1$ and $\delta = k^{\lfloor (n-1)/2 \rfloor}$. Clearly, $k^{\lfloor (n-1)/2 \rfloor} \geq 2n-2$ for $k \geq 6$ and $n \geq 3$, thus, there exists a section, the j th section, contains no faulty rungs. From the symmetric properties of the torus, we can label each vertex (v_2, v_1) in T as $((v_2-(j+0.5)\delta) \bmod k^{n-1}, v_1)$ to generate a torus T_1 that is $TOR(k^{n-1}, k)$ such that the $(v_2, *)$ cycles, in T_1 are all non-faulty, where $0 \leq v_2 \leq 0.5\delta-1$ or $N-0.5\delta \leq v_2 \leq N-1$. The edges in the $(v_2, *)$ cycles of T_1 can be partitioned into two sets $E_o = \{((v_2, v_1), (v_2, v_1+1)) \mid v_1 \text{ is odd}, 0 \leq v_2 \leq k^{n-1}-1, 0 \leq v_1 \leq k-1\}$ and $E_e = \{((v_2, v_1), (v_2, v_1+1)) \mid v_1 \text{ is even}, 0 \leq v_2 \leq k^{n-1}-1, 0 \leq v_1 \leq k-1\}$. Because there are at most $2n-2$ edge faults, there exists a set $(E_o \text{ or } E_e)$ has at most $n-1$ edge faults. Without loss of generality, we assume that E_o has at most $n-1$ edge faults. If we choose E_e to build the ladders: in T_1 , let $BC_1 = (0, *)$ cycle, $LD_1(0) = [(0, 0), (1, 0), \dots, (0.5\delta-2, 0), (0.5\delta-1, 0), (0.5\delta-1, 1), (0.5\delta-2, 1), \dots, (1, 1), (0, 1)]$, $LD_1(1) = [(0, 2), (1, 2), \dots, (N-2, 2), (N-1, 2), (N-1, 3), (N-2, 3), \dots, (1, 3), (0, 3)]$, ..., $LD_1(k/2-1) = [(0, k-2), (1, k-2), \dots, (N-2, k-2), (N-1, k-2), (N-1, k-1), (N-2, k-1), \dots, (1, k-1), (0, k-1)]$; whereas if we choose E_o to build the ladders: in T_1 , let $BC_1 = (0, *)$ cycle, $LD_1(0) = [(0, 1), (1, 1), \dots, (0.5\delta-2, 1), (0.5\delta-1, 1), (0.5\delta-1, 2), (0.5\delta-2, 2), \dots, (1, 2), (0, 2)]$, $LD_1(1) = [(0, 3), (1, 3), \dots, (N-2, 3), (N-1, 3), (N-1, 4), (N-2, 4), \dots, (1, 4), (0, 4)]$, ..., $LD_1(k/2-1) = [(0, k-1), (1, k-1), \dots, (N-2, k-1), (N-1, k-1), (N-1, 0),$

$(N-2, 0), \dots, (1, 0), (0, 0)]$. Unifying the bone cycle BC_1 with the $k/2$ ladders, $col_1 = LWCOL(BC_1, k/2, LD_1(0), LD_1(1), \dots, LD_1(k/2-1))$ can be derived, where $LD_1(0)$ with the length $0.5\delta-1$ has no faulty rungs. Clearly, $0.5\delta-1 = 0.5k^{\lceil (n-1)/2 \rceil} - 1 \geq n-1$ for $k \geq 6$ and $n \geq 3$. From Lemma 1, we know that there exists a cycle of each even length ranging from k to $k^n - 2(k^{n-1} - 0.5\delta) = k^n - 2k^{n-1} + \delta$ that is the order of the $LWCOL(BC_1, k/2, LD_1(0), LD_1(1), \dots, LD_1(k/2-1))$. Let BC_2 contain all the vertices of col_1 except the $(N-1)$ th rung, the $(N-2)$ th rung, $\dots$, the $(N-0.5\delta+2)$ th rung and the $(N-0.5\delta+1)$ th rung of the $LD_1(1)$. Let $LD_2(0) = [(0.5\delta-1, 0), (0.5\delta, 0), \dots, (N-2, 0), (N-1, 0), (N-1, 1), (N-2, 1), \dots, (0.5\delta, 1), (0.5\delta-1, 1)]$, $LD_2(1) = [(N-0.5\delta, 2), (N-0.5\delta+1, 2), \dots, (N-2, 2), (N-1, 2), (N-1, 3), (N-2, 3), \dots, (N-0.5\delta+1, 3), (N-0.5\delta, 3)]$. Unifying the bone cycle BC_2 with the 2 ladders, $col_2 = LWCOL(BC_2, 2, LD_2(0), LD_2(1))$ can be derived, where $LD_2(1)$ with the length $0.5\delta-1$ has no faulty rungs. Clearly, $0.5\delta-1 = 0.5k^{\lceil (n-1)/2 \rceil} - 1 \geq n-1$ for $k \geq 6$ and $n \geq 3$. From Lemma 1, we know that there exists a cycle of each even length ranging from $k^n - 2(k^{n-1} - 0.5\delta) - 2(0.5\delta-1) = k^n - 2k^{n-1} + 2$ to k^n . Clearly, $k^n - 2k^{n-1} + 2 \leq k^n - 2k^{n-1} + \delta$ for $k \geq 6$ and $n \geq 3$; thus, combining the two results, there exists a cycle of each even length ranging from k to k^n in even $Q(k, n)$ with at most $2n-2$ edge faults for $k \geq 6$ and $n \geq 3$. Q. E. D.

For example, consider a $Q(6, 5)$ with $2n-2 = 8$ edge faults. firstly, we show that the $Q(6, 5)$ embed a $TOR(6^4, 6)$ contains at most 8 edge faults, where the cycles $(*, v_1)$ contain no edge faults, where $0 \leq v_1 \leq 5$. From the symmetric properties of the $Q(k, n)$, we assume that $\lambda(5) \leq \lambda(4) \leq \lambda(3) \leq \lambda(2) \leq \lambda(1)$, where the number of faulty edges in dimension i is denoted by $\lambda(i)$. Clearly, $2 \leq \lambda(1) \leq 8$; thus, the $OG(Q(6, 5))$ is a $Q(6, 4)$ with at most 6 edge faults. From Proposition 4.1, there exists a Hamiltonian cycle in the $OG(Q(6, 5))$. From Proposition 6, the $Q(6, 5)$ embeds a torus T that is a $TOR(6^4, 6)$ contains at most 8 edge faults, where the cycles $(*, v_1)$ contain no edge faults, where $0 \leq v_1 \leq 5$.

Then, we show that T contains a cycle of each even length ranging from 4 to 6^5 . For example, as illustrated in Figure 4, consider $TOR(6^4, 6)$ with eight edge faults, $((1, 0), (1, 1)), ((2, 0), (2, 1)), ((71, 4), (71, 5)), ((238, 3), (238, 4)), ((396, 2), (396, 3)), ((658, 5), (658, 0)), ((870, 1), (870, 2)), ((1294, 3), (1294, 4))$. From the $(0, *)$ cycle, we can partition the 6^4 $(v_2, *)$ cycles into $6^{\lceil (5-1)/2 \rceil} = 36$ sections in which each section contains $\delta = 6^{\lceil (5-1)/2 \rceil} = 36$ successive $(v_2, *)$ cycles, where $0 \leq v_2 \leq 6^4-1 = 1295$. That is, the i th section contains $(36i, *)$, $(36i+1, *)$, $\dots$, $(36i+34, *)$ and $(36i+35, *)$, where

$0 \leq i \leq 35$. The edge faults, $((1, 0), (1, 1)), ((2, 0), (2, 1)), ((71, 4), (71, 5)), ((238, 3), (238, 4)), ((396, 2), (396, 3)), ((658, 5), (658, 0)), ((870, 1), (870, 2)), ((1294, 3), (1294, 4))$, are contained in the 0th section, the 0th section, the 1st section, the 6th section, the 11th section, the 18th section, the 24th section and the 35th section, respectively. Because the number of edge faults is less than the number of sections, we can find a section without any edge fault. For example, the 2nd section has no edge faults. Thus, T embeds the ladder $LD(0) = [(j\delta = 72, 0), (73, 0), \dots, (106, 0), (107, 0), ((107, 1), (106, 1), \dots, (73, 1), (72, 1))]$. From Proposition 1, $LD(0)$ embeds a cycle of each length ranging from 4 to $2\delta = 72$ that is greater than $k = 6$ for $n \geq 3$. Thus, we can label each vertex (v_2, v_1) in T as $(u_2, v_1) = ((v_2 - 2.5\delta = v_2 - 90) \bmod 6^4, v_1)$ in T_1 that is $TOR(6^4, 6)$ in which each $(u_2, *)$ cycle is non-faulty, where $0 \leq u_2 \leq 17$ or $6^4-18 \leq u_2 \leq 6^4-1$. The edge faults will be labeled as $((1207, 0), (1207, 1)), ((1208, 0), (1208, 1)), ((1277, 4), (1277, 5)), ((148, 3), (148, 4)), ((306, 2), (306, 3)), ((568, 5), (568, 6)), ((780, 1), (780, 2))$ and $((1204, 3), (1204, 4))$ in T_1 , respectively. The edges in the $(u_2, *)$ cycles of T_1 can be partitioned into two sets $E_o = \{((u_2, v_1), (u_2, v_1+1)) \mid v_1 \text{ is odd}, 0 \leq u_2 \leq 6^4-1, 0 \leq v_1 \leq 5\}$ and $E_e = \{((u_2, v_1), (u_2, v_1+1)) \mid v_1 \text{ is even}, 0 \leq u_2 \leq 6^4-1, 0 \leq v_1 \leq 5\}$. Because there are at most 8 edge faults, one of the two sets (E_o and E_e) has at most 4 edge faults as illustrated in Figure 5. In this case, both of E_o and E_e have 4 edge faults. If we choose E_e to build the ladders: In T_1 , let $BC_1 = (0, *)$ cycle, $LD_1(0) = [(0, 0), (1, 0), \dots, (16, 0), (17, 0), (17, 1), (16, 1), \dots, (1, 1), (0, 1)]$ with no faulty rungs, $LD_1(1) = [(0, 2), (1, 2), \dots, (6^4-2, 2), (6^4-1, 2), (6^4-1, 3), (6^4-2, 3), \dots, (1, 3), (0, 3)]$ with a faulty rung $((306, 2), (306, 3))$ and $LD_1(2) = [(0, 4), (1, 4), \dots, (6^4-2, 4), (6^4-1, 4), (6^4-1, 5), (6^4-2, 5), \dots, (1, 5), (0, 5)]$ with a faulty rung $((1277, 4), (1277, 5))$. Unifying the bone cycle BC_1 with the 3 ladders, $LWCOL(BC_1, 3, LD_1(0), LD_1(1), LD_1(2))$ with $LWCOL$ block length 1 can be derived. As illustrated in Figure 6, the case for embedding $LWCOL(BC_1, 3, LD_1(0), LD_1(1), LD_1(2))$ into T_1 is shown. Since the $LWCOL$ block length is less than 17, from Lemma 1, we know that there exists a cycle of each even length ranging from $k = 6$ to $k^n - 2k^{n-1} + \delta = 7776 - 2592 + 36 = 5220$ that is the order of the $LWCOL(BC_1, 3, LD_1(0), LD_1(1), LD_1(2))$. Let BC_2 contain all the vertices of col_1 except the (6^4-1) th rung, the (6^4-2) th rung, $\dots$, the (6^4-16) th rung and the (6^4-17) th rung of the $LD_1(1)$. That is, $BC_2 = ((0, 0), (1, 0), \dots, (16, 0), (17, 0), (17, 1), (16, 1), \dots, (1, 1), (0, 1), (0, 2), (1, 2), \dots, (6^4-19, 2), (6^4-18, 2), (6^4-18, 3), (6^4-19, 3), \dots, (1, 3), (0, 3), (0, 4), (1, 4), \dots, (6^4-2, 4), (6^4-1, 4), (6^4-1, 5), (6^4-2, 5), \dots, (1, 5), (0, 5), (0, 0))$. Let $LD_2(0) = [(17, 0), (18, 0), \dots, (6^4-2, 0), (6^4-1, 0), (6^4-1, 1),$

$(6^4-2, 1), \dots, (18, 1), (17, 1)]$ with two faulty rungs $((1207, 0), (1207, 1))$ and $((1208, 0), (1208, 1))$, $LD_2(1) = [(6^4-18, 2), (6^4-17, 2), \dots, (6^4-2, 2), (6^4-1, 2), (6^4-1, 3), (6^4-2, 3), \dots, (6^4-17, 3), (6^4-18, 3)]$ with no faulty rungs. Unifying the bone cycle BC_2 with the 2 ladders, $col_2 = LWCOL(BC_2, 2, LD_2(0), LD_2(1))$ with LWCOL block length 2 can be derived, where $LD_2(1)$ with the length $0.5\delta-1 = 17$ has no faulty rungs. As illustrated in Figure 7, the case for embedding $LWCOL(BC_2, 2, LD_2(0), LD_2(1))$ into T_1 is shown. Since the LWCOL block length of col_2 is 2, from Lemma 1, we know that there exists a cycle of each even length ranging from $k^n-2k^{n-1}+2 = 7776-2592+2 = 5186$ to $k^n = 7776$. Combining the above results, there exists a cycle of each even length ranging from 4 to $6^5 = 7776$.

Combining Lemma 2 and Lemma 3, we know:

Lemma 4. The even $Q(k, n)$ is $(2n-2)$ -edge-pancyclic for $k \geq 6$ and $n \geq 2$.

Because $Q(2, n)$ is an n -dimensional binary hypercube and $Q(4, n)$ is an $2n$ -dimensional binary hypercube, from Proposition 4 and Lemma 4, we can state:

Theorem 5. The even $Q(k, n)$ is $(2n-2)$ -edge-pancyclic for $n \geq 2$.

References

- [1] Ametek Corporation, Ametek Series 2010 Brochures.
- [2] T. Araki and Y. Shibata, Pancyclicity of recursive circulant graphs, *Information Processing Letters* 81(4) (2002) 187-190.
- [3] Y. Ashir, I.A. Stewart, A. Ahmed, Communication algorithms in k -ary n -cube interconnection networks, *Information Processing Letters*, 61(1) (1997) 43-48.
- [4] B. Bose, B. Broeg, Y. Kwon, and Y. Ashir, "Lee Distance and Topological Properties of k -Ary n -Cubes", *IEEE Trans. Computers*, 1995, 1021-1030.
- [5] J. Bruck, R. Cypher, C.T. Ho, Wildcard dimensions, coding theory and fault-tolerant meshes and hypercubes, *IEEE Transaction on Computers* 44(1) (1995) 150-155.
- [6] M.Y. Chan, F.Y.L. Chin, C.K. Poon, Optimal simulation of full binary trees on faulty hypercubes, *IEEE Transaction on Parallel and Distributed Systems* 6(3) (1995) 269-286.
- [7] A.H. Esfahanian, S.L. Hakimi, Fault-tolerant routing in de Bruijn communication networks, *IEEE Transactions on Computers* C-34 (9) (1985) 777-788.
- [8] J.F. Fang, Y.R. Wang and H.L. Huang, The m -Pancycle connectivity of a WK-Recursive network, *Information Sciences* 177 (24) (2007) 5611-5619.
- [9] S.Y. Hsieh and C.H. Chen, Pancyclicity on Möbius cubes with maximal edge faults, *Parallel Computing* 30(3) (2004) 407-421.
- [10] S. Lakshmivarahan, S.K. Dhall, Ring, torus and hypercube architectures/algorithms for parallel computing, *Parallel Computing* 25 (1999) 1877-1906.
- [11] F.T. Leighton, *Introduction to Parallel Algorithms and Architectures: Arrays, Trees, Hypercubes*, Morgan Kaufmann, California, 1992.
- [12] T.K. Li, Cycle embedding in star graphs with edge faults, *Applied Mathematics and Computation* 167(2) (2005) 891-900.
- [13] Y.C. Lin, On balancing sorting on a linear array, *IEEE Transactions on Parallel and Distributed Systems* 4(5) (1993) 566-571.
- [14] NCUBE Corporation, *NCUBE Handbook*, Beaverton, Ore., (1986).
- [15] W. Oed, *Massively Parallel Processor System CRAY T3D*, technical report, Cray Research GmbH, (1993).
- [16] D.R. O'Hallaron, Uniform approach for solving some classical problems on a linear array, *IEEE Transactions on Parallel and Distributed Systems* 2(2) (1991) 236-241.
- [17] C.L. Seitz, The cosmic cube, *Comm. ACM*, 28(1) (1985) 22-33.
- [18] Tera Computer Systems, *Overview of the Tera Parallel Computer*, (1993).
- [19] J.M. Xu, Z.Z. Du, M. Xu, Edge-fault-tolerant edge-bipancyclicity of hypercubes, *Information Processing Letters* 96 (2005) 146-150.
- [20] J.M. Xu, M. Ma, Cycles in folded hypercubes, *Applied Mathematics Letters* 19(2) (2006) 140-145.
- [21] M.C. Yang, J.J.M. Tan, L.H. Hsu, Hamiltonian circuit and linear array embeddings in faulty k -ary n -cubes, *Journal of Parallel and Distributed Computing*, 67(4) (2007) 362-368.
- [22] T.K. Li, C.H. Tsai, J.J.M. Tan and L.H. Hsu, Bipanconnectivity and edge-fault-tolerant bipancyclicity of hypercubes, *Information Processing Letters* 87 (2003) 107-110.

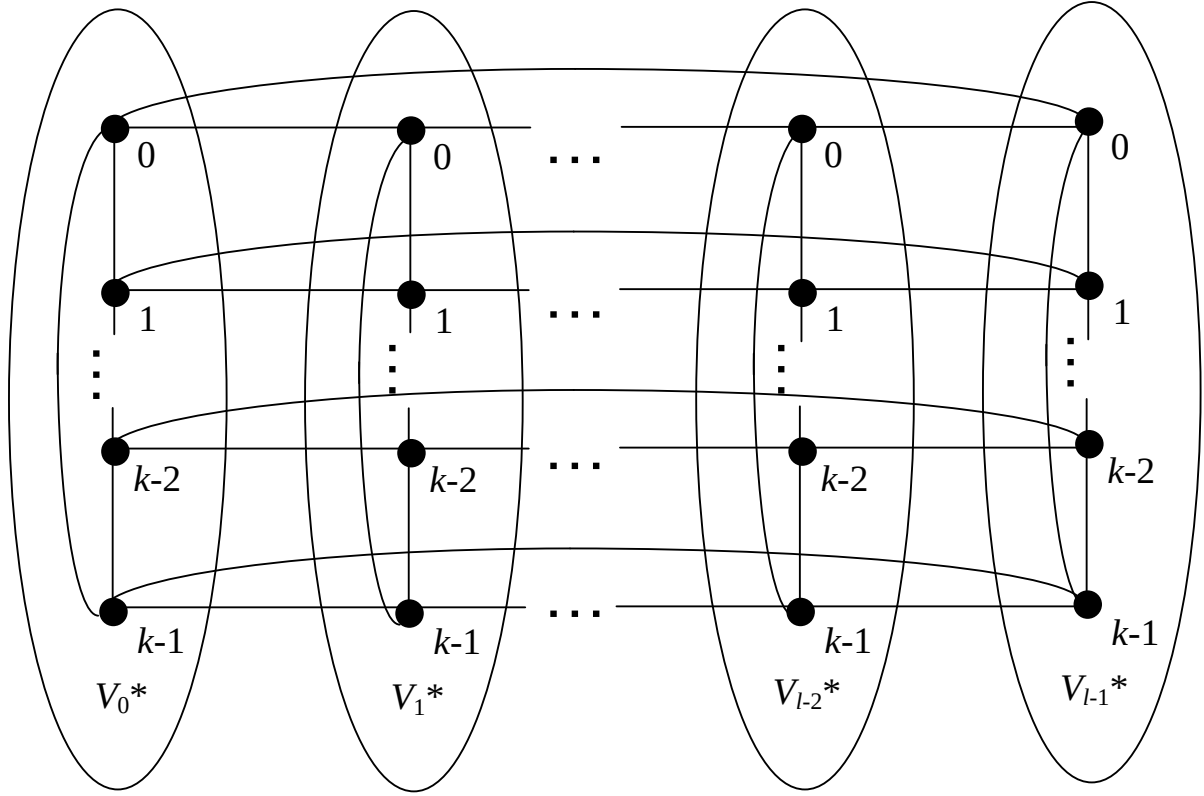


Figure 1. Embedding the structure of $C_l \times C_k$ into the $Q(k, n)$ if $C_l, (V_0^*, V_1^*, V_2^*, \dots, V_{l-1}^*, V_0^*)$, is embedded in the $OG(Q(k, n))$.

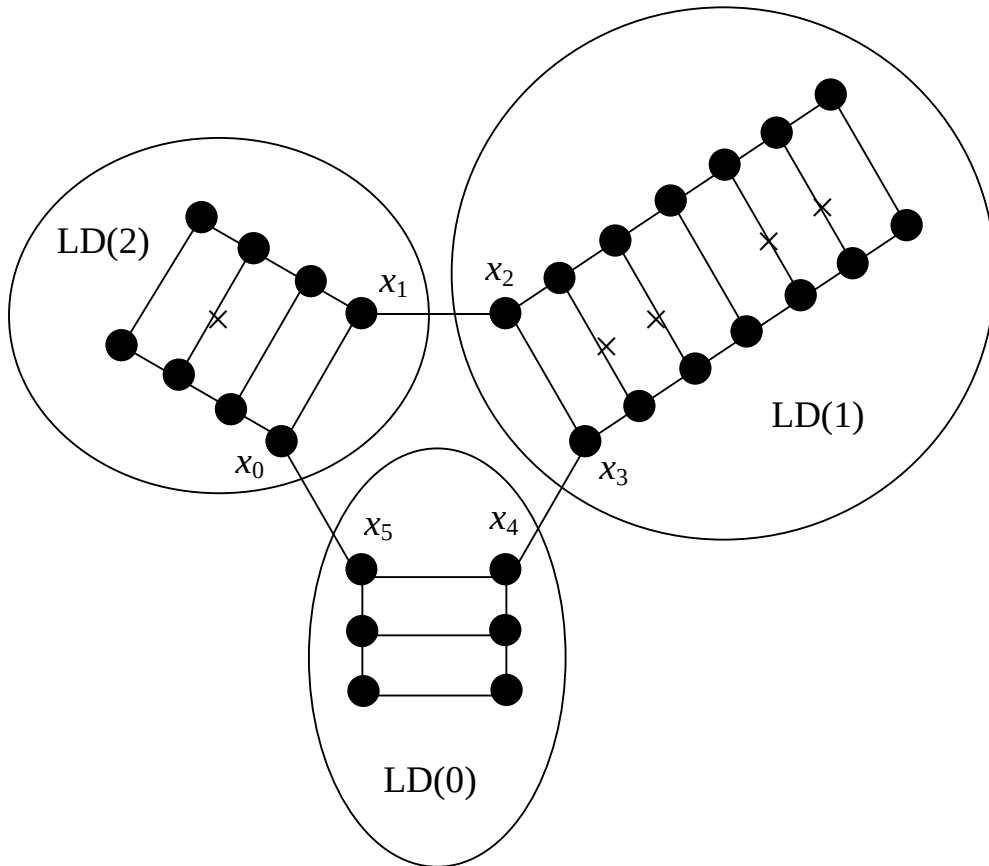


Figure 2. An example of $LWCOL(BC_1, 3, LD(0), LD(1), LD(2))$ with one non-faulty ladder $LD(0)$ of the length 2 and the LWCOL block length $f_s = 2$, where $BC = (x_0, x_1, x_2, x_3, x_4, x_5, x_0)$.

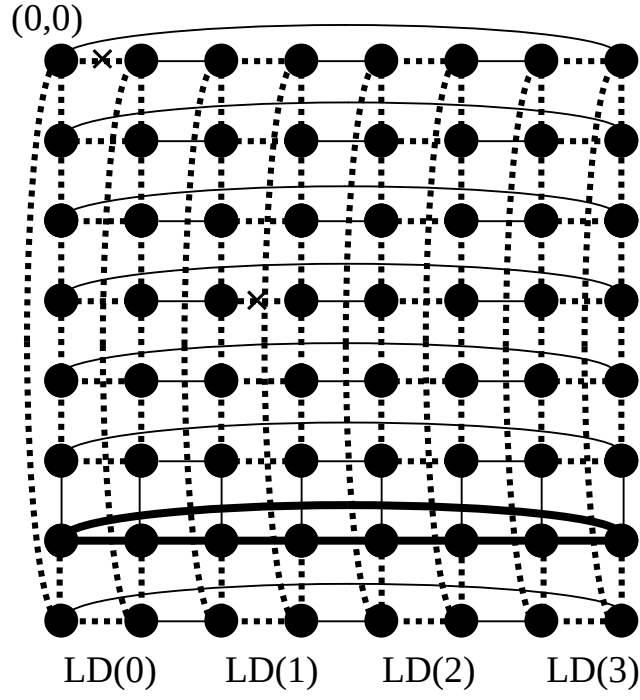


Figure 3. Generating the cycle-of-ladders for the case that $(y_2, y_1) = ((v_2, v_1), (v_2, v_1+1)) = ((3, 2), (3, 3))$ and $k = 8$.

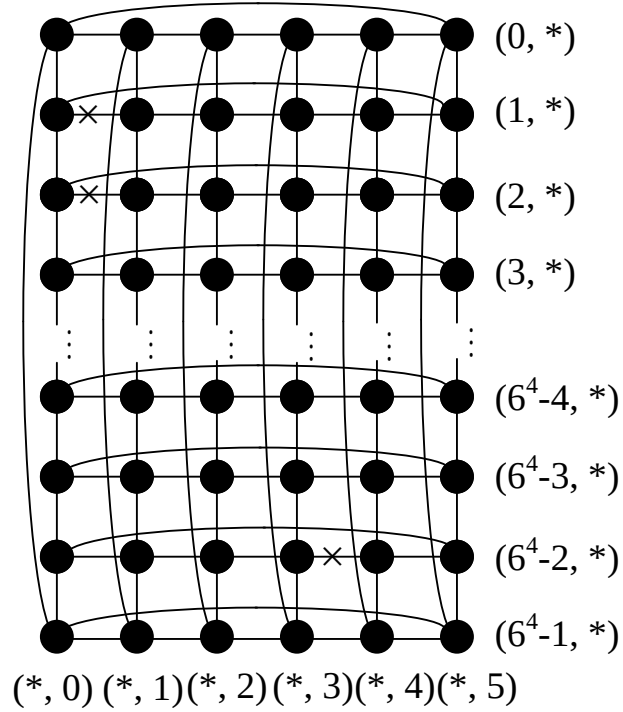


Figure 4. An example of $\text{TOR}(6^4, 6)$ with 8 edge faults.

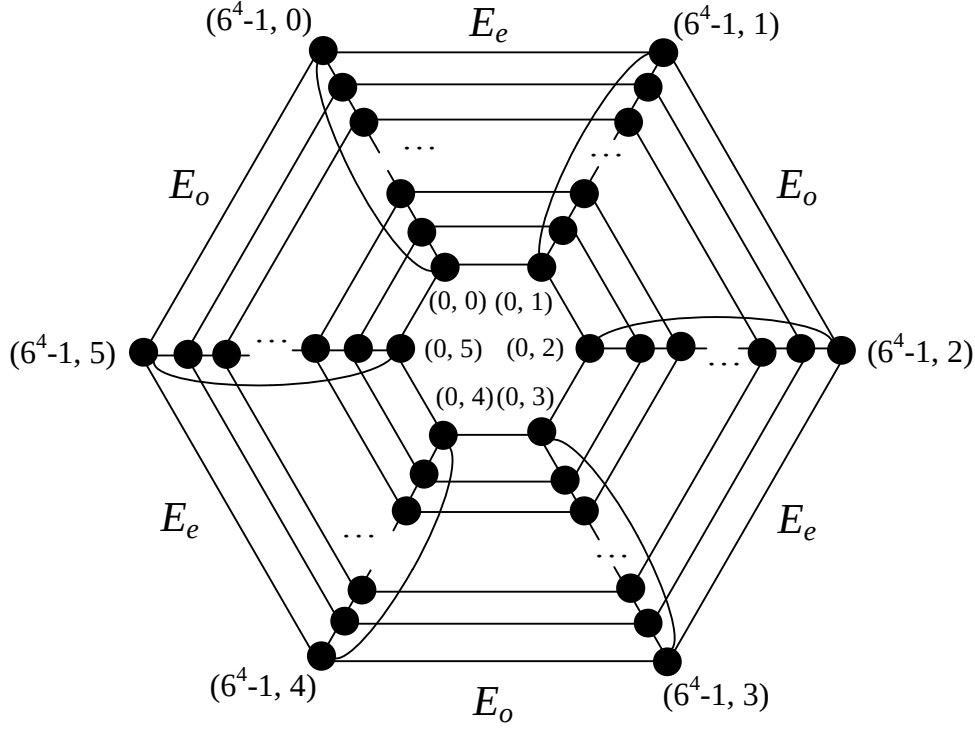


Figure 5. One of E_e or E_o has at most $n-1 = 4$ edge faults, where $E_e = \{((1207, 0), (1207, 1)), ((1208, 0), (1208, 1)), ((306, 2), (306, 3)), ((1277, 4), (1277, 5))\}$, $E_o = \{((780, 1), (780, 2)), ((148, 3), (148, 4)), ((1204, 3), (1204, 4)), ((568, 5), (568, 6))\}$.

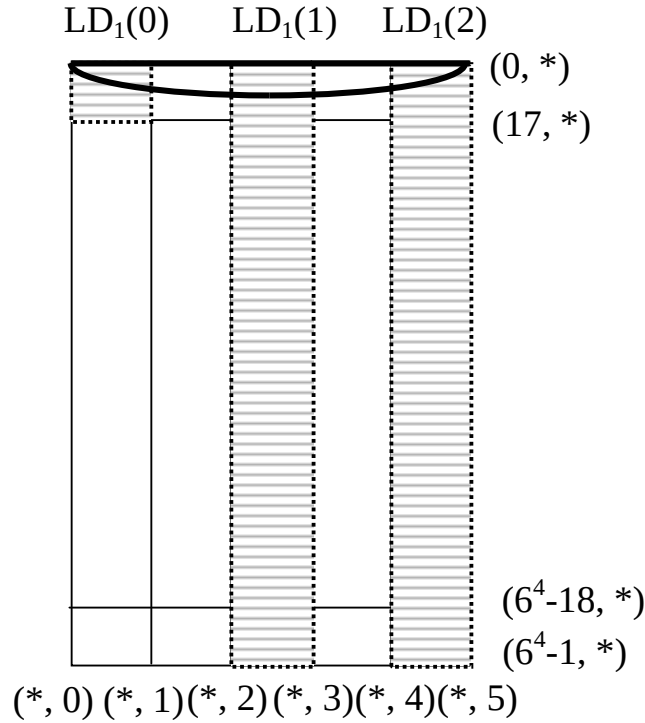


Figure 6. Embedding $LWCOL(BC_1, 3, LD_1(0), LD_1(1), LD_1(2))$ into T_1 , where the thick line is the bone cycle BC_1 and the dotted line is the ladders.

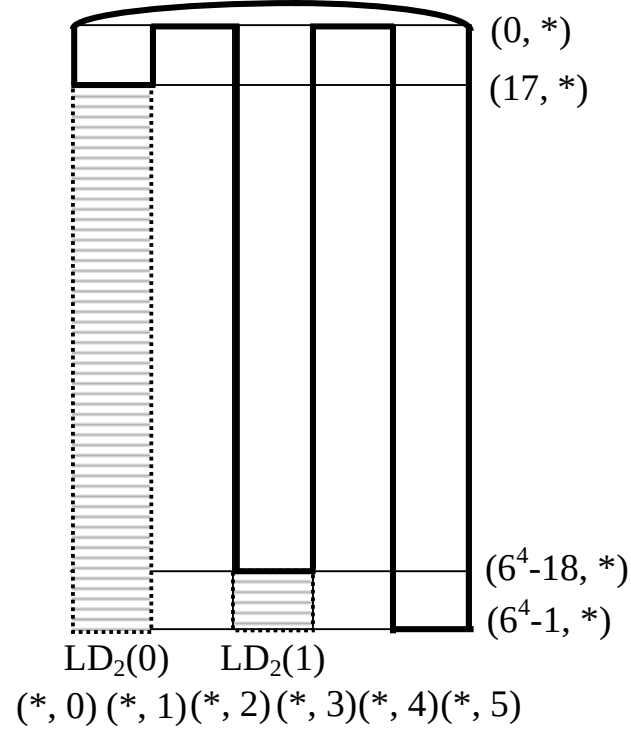


Figure 7. Embedding $LWCOL(BC_2, 2, LD_2(0), LD_2(1))$ into T_1 , where the thick line is the bone cycle BC_2 and the dotted line is the ladders.